

Mining Patterns from Large-Scale Flight Data

Feature Engineering and Classification for Delay Prediction at ATL

Yuntian Robin Wu
Joshua Pina

The Data Challenge

Big Data in Aviation

Modern aviation generates massive operational data volumes. Flight schedules, weather, sensors, air traffic control—hidden patterns require systematic mining.

Scale of the Problem

4+ million flight records annually with 109 attributes per flight. Complex interdependencies, missing data, and need for scalable workflows.

Project Objectives

Knowledge Discovery Goals

- Extract actionable insights from historical flight data
- Develop predictive models for operational decisions
- Build systematic preprocessing pipeline
- Validate on world's busiest airport

Success Criteria

- High prediction accuracy on test data
- Balanced performance across delay types
- Scalable to real-world deployment
- Interpretable and explainable results

Data Source & Scope

4M

Flight Records

US Bureau of
Transportation
Statistics 2022 data
covering domestic
flights nationwide

109

Features

Attributes covering
operations, timing,
airlines, and airports

184K

ATL Flights

Focus on world's
busiest airport with
high traffic volume and
diverse airline mix

Initial Data Quality Assessment

Duplicate Records

12,847 duplicate records identified requiring removal to prevent training bias

Missing Values

23 features with 100% missing values and varying levels across other features

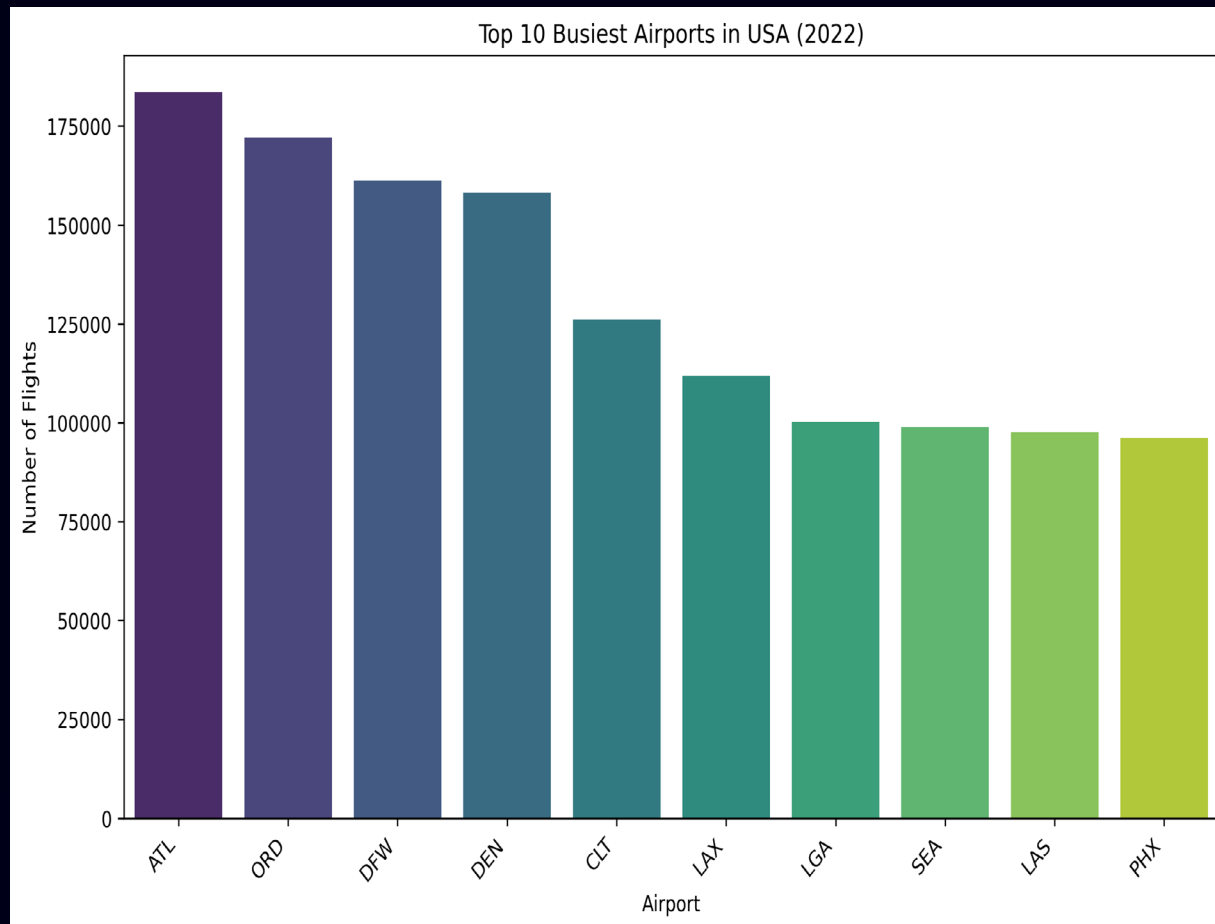
Format Issues

Inconsistent time formats (HHMM encoding) requiring standardization

Class Imbalance

80% on-time vs 20% delayed flights affecting model training

Data quality determines model quality—comprehensive cleaning pipeline required before applying algorithms.



Traffic Distribution Analysis

Top 10 airports handle 28% of all flights, with ATL leading at 183K flights. Significant concentration in major hubs with long tail of smaller airports.

Systematic Cleaning Steps

01

Remove Duplicates

Eliminated 12,847 duplicate records to prevent training bias and ensure data integrity

03

Eliminate ID Fields

Identified and removed 27 ID/unavailable features that cause overfitting

02

Drop Null Features

Removed 23 features with 100% missing values that provide no information

04

Feature Reduction

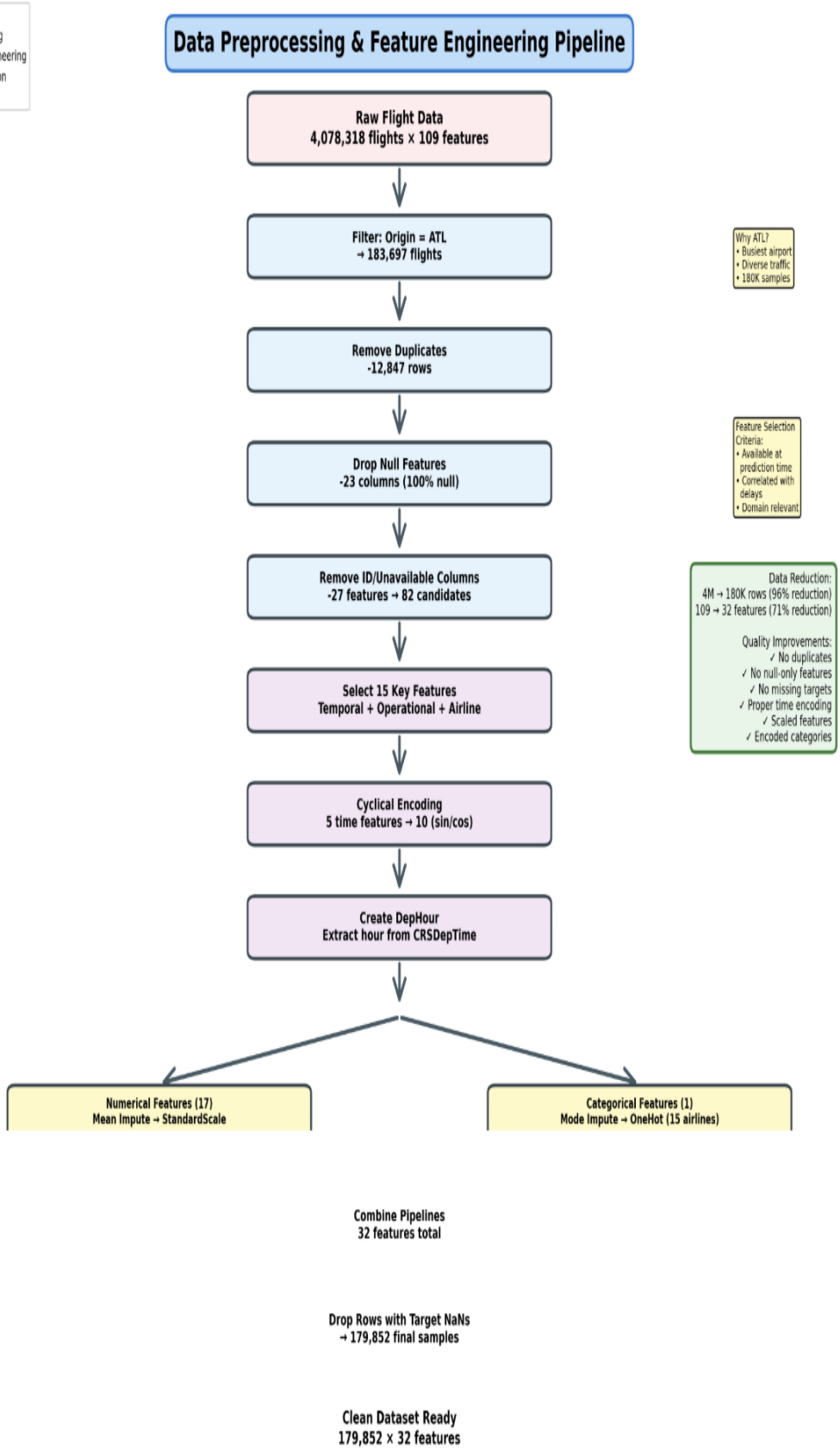
Reduced from 109 to 82 candidate features for further analysis



DATA
CLEANING

FEATURE
ENGINEERING

TRANSFORM



Feature Selection Process

Selection Criteria

- Available at prediction time
- Relevant to delays via domain knowledge
- Correlated with target variable
- Avoid unique identifiers

From 82 to 15 Features

Temporal: Month, DayofMonth, DepHour

Operational: TaxiIn, TaxiOut, AirTime, Distance

Scheduled/Actual Times: 5 time features

Delay Indicator: DepDel15

Categorical: Airline

Missing Value Handling

1

Missing Data Patterns

Different patterns across feature types requiring strategic approach

2

Imputation Strategy

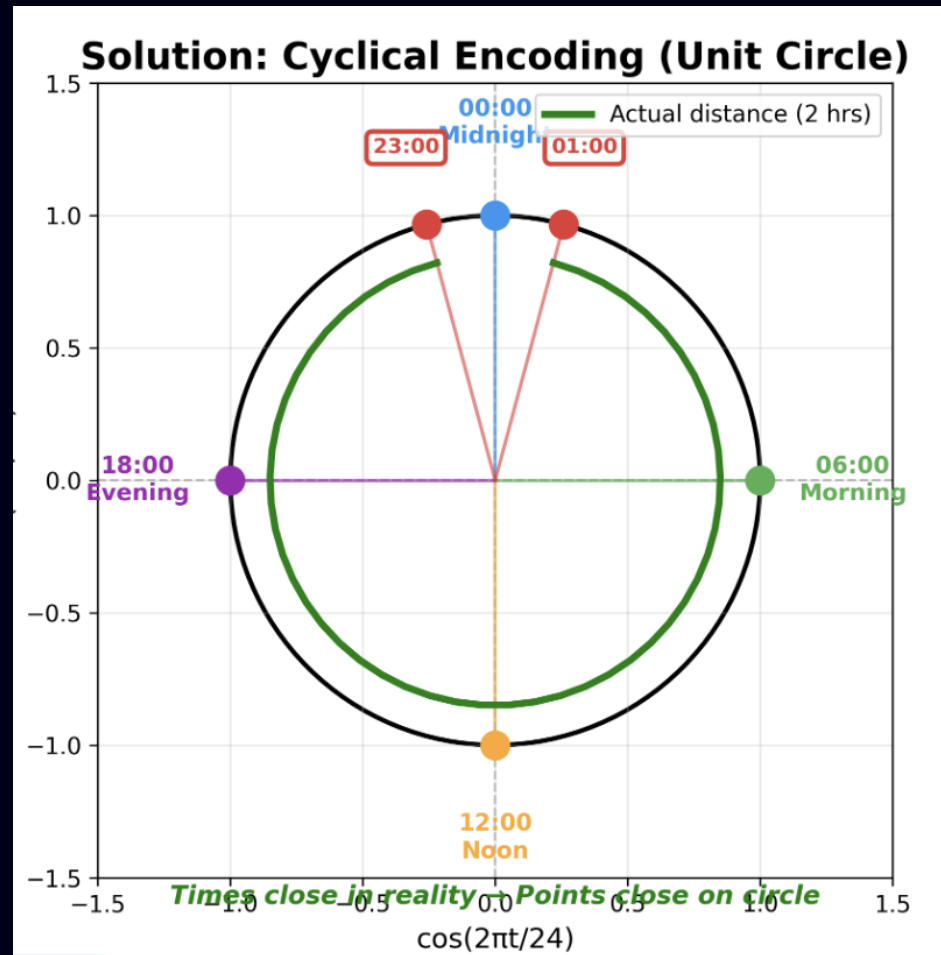
Numerical: mean imputation. Categorical: most frequent.
Target: row deletion

3

Two-Stage Approach

Select features first, then impute to preserve statistical properties

The Temporal Feature Challenge



Problem with Time Data

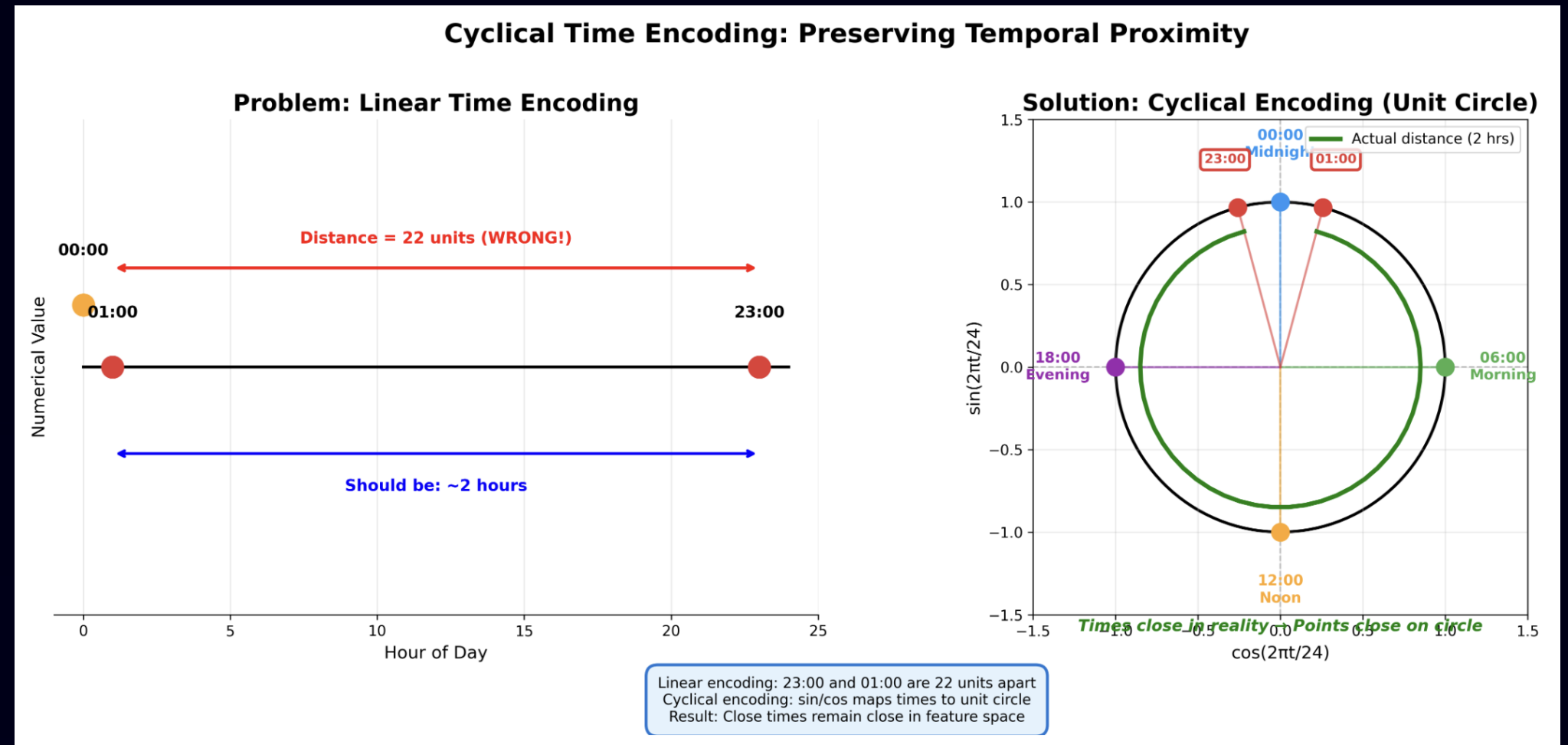
Time encoded as integers (HHMM format). 23:59 and 00:01 are numerically distant (2358 units) but temporally adjacent.

Linear distance doesn't reflect temporal proximity—models learn incorrect patterns.

Real-World Impact

- Cannot capture late night/early morning similarities
- Misses circadian patterns in operations
- Artificially breaks temporal continuity
- Reduces prediction accuracy

Cyclical Encoding Solution



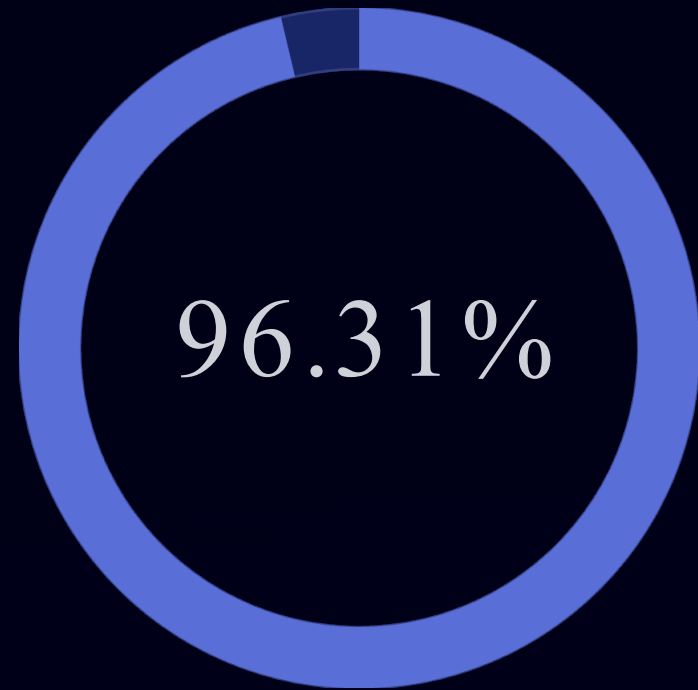
Geometric Interpretation

Maps 24-hour cycle onto unit circle. Midnight at (0,1), noon at (0,-1). Adjacent times map to nearby points.

Convert HHMM to minutes: $m = (t \div 100) \times 60 + (t \bmod 100)$

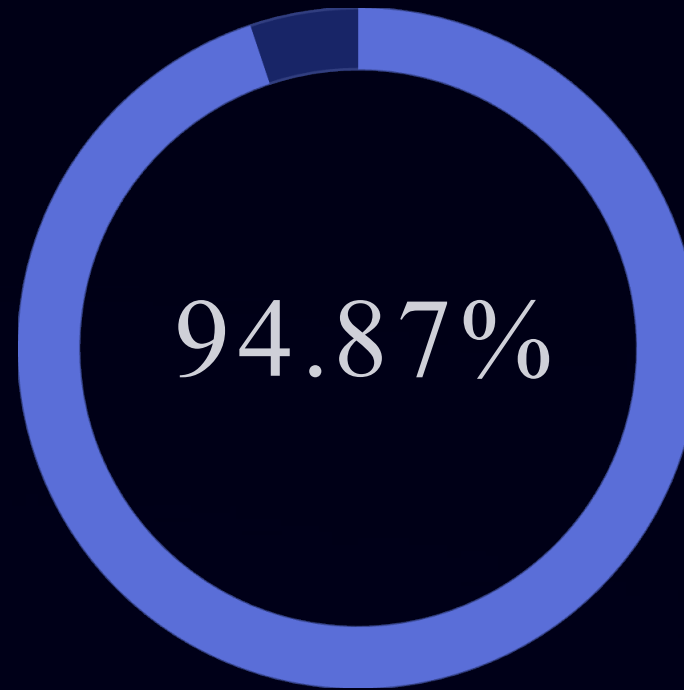
Apply $\sin(2\pi m/1440)$ and $\cos(2\pi m/1440)$ creating two features per time dimension

Impact Quantification



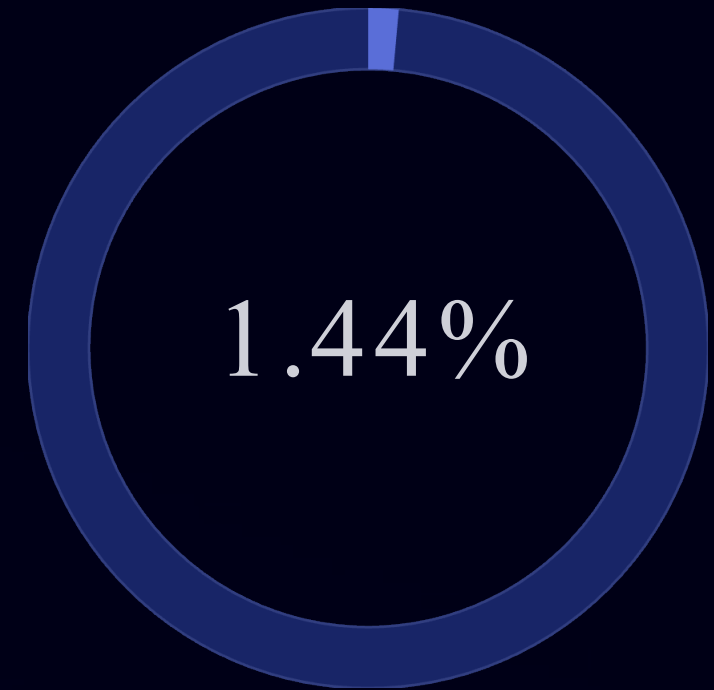
With Cyclical Encoding

Accuracy achieved using proper temporal representation



Without Cyclical Encoding

Baseline accuracy with raw HHMM format

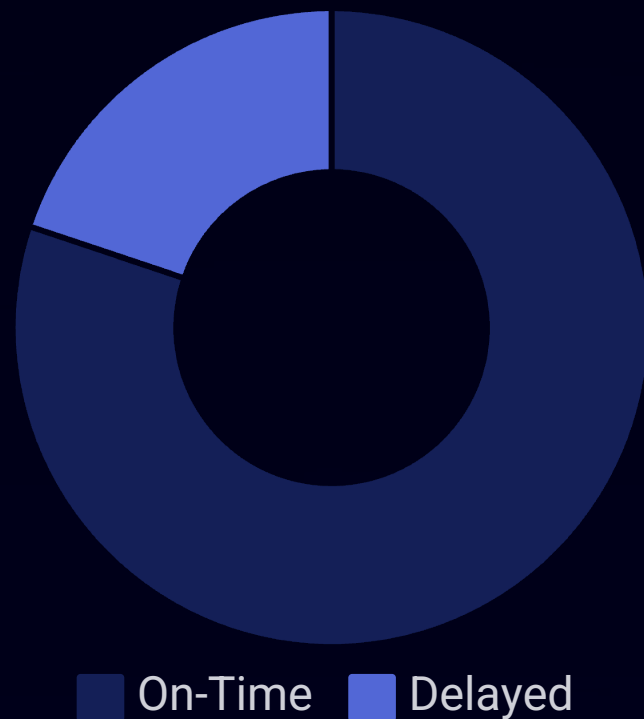


Improvement

~520 additional correct predictions on test set—largest single preprocessing contribution

Validates feature engineering effort and demonstrates importance of domain-appropriate encoding.

Class Imbalance Characteristics



Distribution Analysis

On-time flights: 144,410 (80.3%)

Delayed flights: 35,442 (19.7%)

Ratio: approximately 4:1

Implications

Classifiers may bias toward majority class. Standard accuracy can be misleading—need class-specific metrics.

Handling Strategies Considered

Oversampling

SMOTE, ADASYN to increase minority class representation

Undersampling

Reduce majority class to balance distribution

Cost-Sensitive Learning

Apply class weights to penalize errors differently

Natural Distribution

Our choice: Keep natural distribution, use robust algorithms and class-specific metrics

Results Validation

97%

On-Time Precision

98% recall for on-time flights

92%

Delayed Precision

89% recall for delayed flights

0.94

Macro F1

Average across classes

0.96

Weighted F1

Accounting for class size

Key Takeaway: Random Forest's ensemble structure naturally handles 4:1 imbalance. No explicit resampling needed—multiple trees capture minority patterns effectively.

Algorithm Selection



Random Forest

Ensemble method building multiple decision trees on bootstrap samples. Excels at capturing complex patterns and feature interactions while reducing overfitting through aggregation.



Logistic Regression

Linear baseline model providing interpretable coefficients. Establishes performance floor and helps identify when complex models add genuine value.



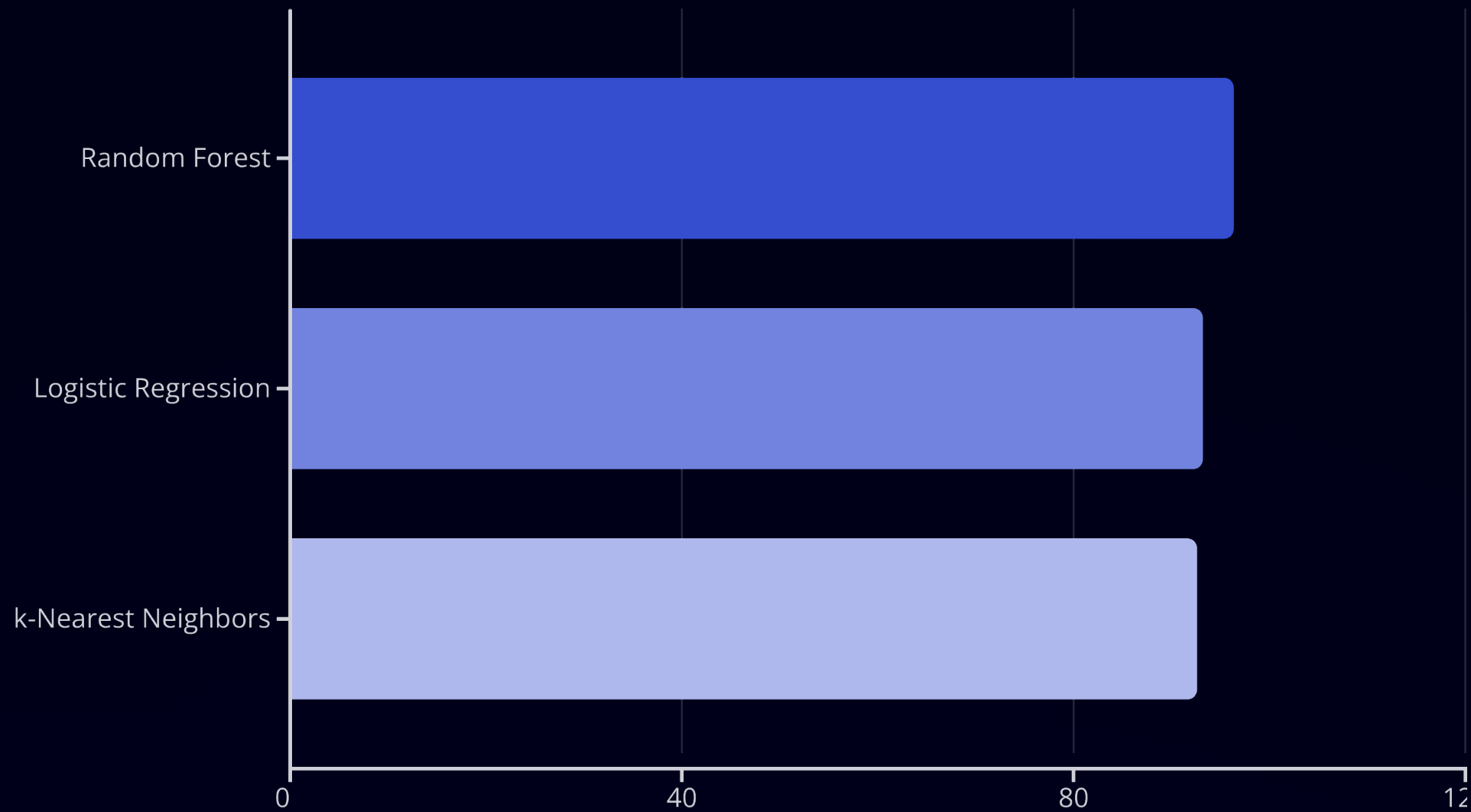
k- Nearest Neighbors

Non-parametric approach for local pattern matching. Useful for identifying region-specific delay behaviors without assuming global functional form.

These three algorithms represent fundamentally different learning paradigms, enabling comprehensive performance comparison across linear, ensemble, and instance-based methods.

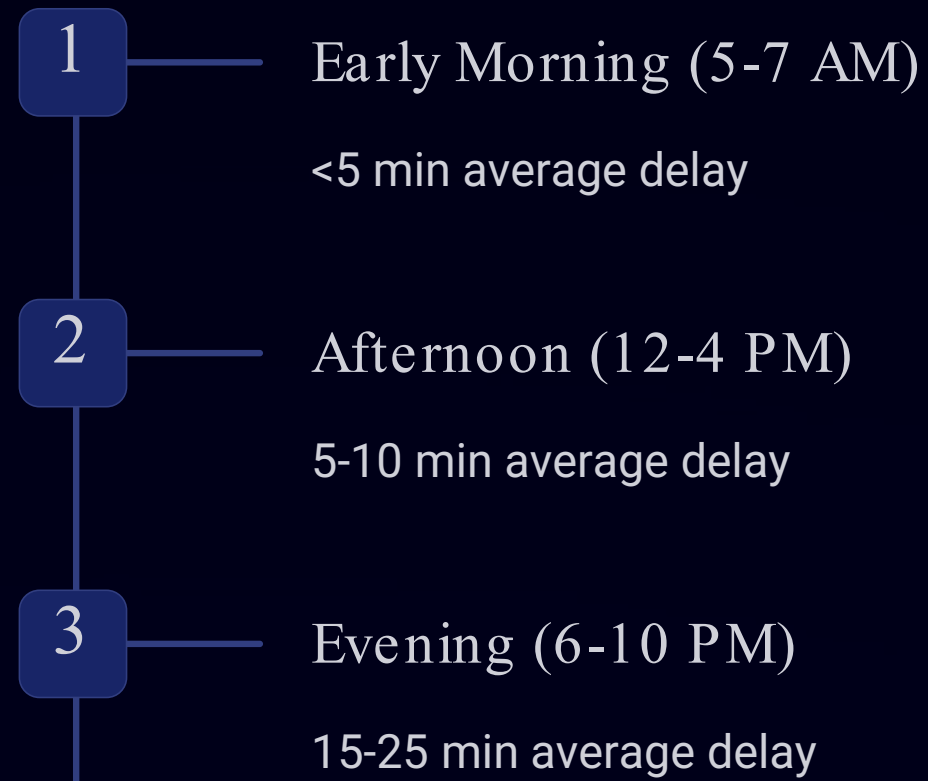
Performance Results

Clear winner: Random Forest with 3+ point advantage over alternatives



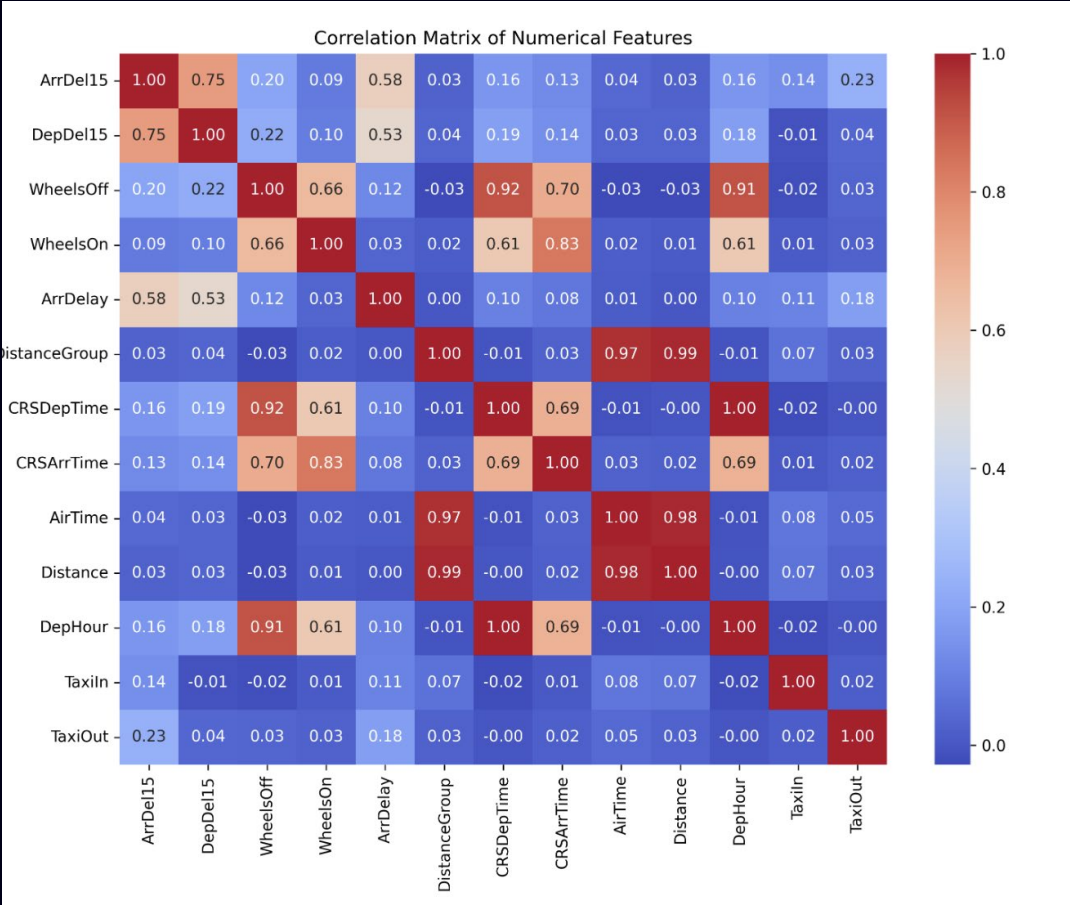
3% difference equals 1,100 correct predictions on 36K test set. Consistent across multiple metrics and robust to different random seeds.

Temporal Patterns Discovered



Explanation: Cascading delay effect throughout daily operations. Same aircraft/crew operate multiple flights—morning delays propagate through schedule with network effects compounding over time.

Feature Correlation Patterns



Strong Correlations

DepDel15 ↔ ArrDel15: 0.93 (departure delays predict arrival)

Distance ↔ DistanceGroup: 0.99 (redundant features)

- Moderate correlations among time features

Weak Correlations

Distance ↔ Delays: weak correlation

- Distance alone not predictive
- Other factors dominate delay patterns
- Validates multi-feature approach

Confusion Matrix Deep Dive

True Negatives

28,236 correctly predicted on-time flights (strong majority class performance)

True Positives

6,371 correctly predicted delays (good minority detection)

False Positives

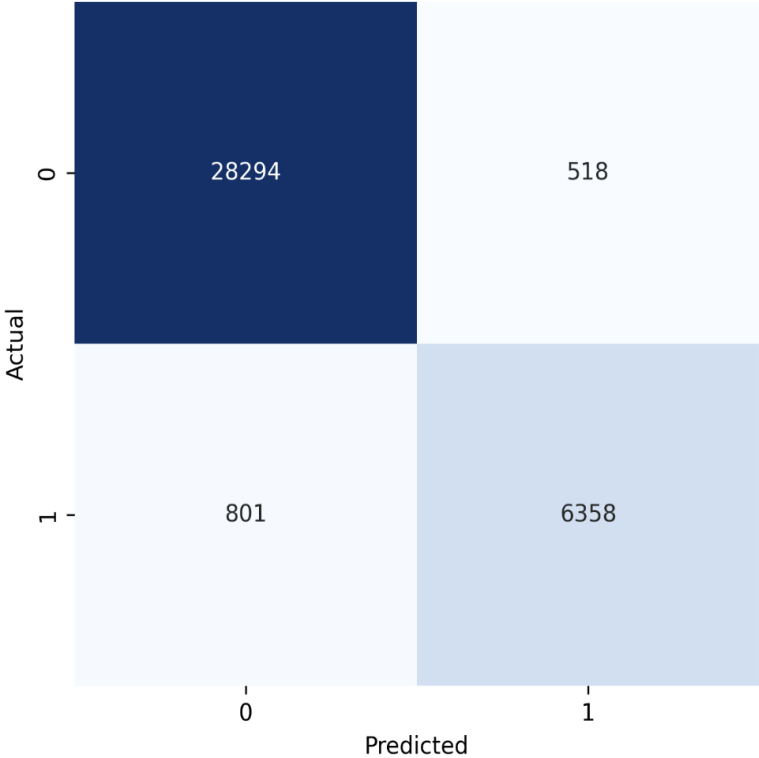
576 incorrect delay predictions (2% false alarm rate)

False Negatives

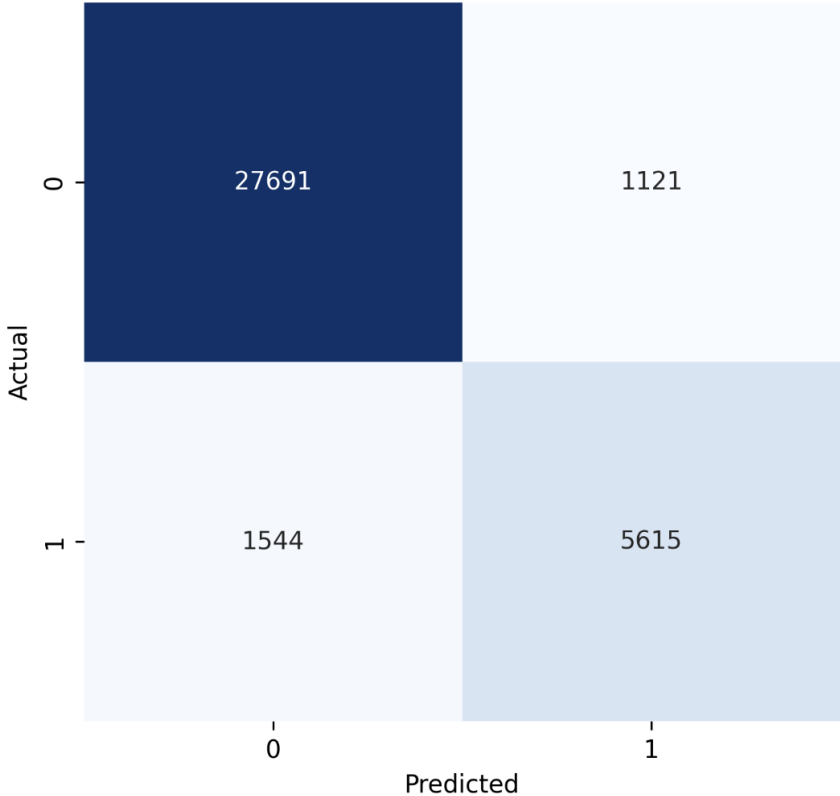
788 missed delays (11% miss rate)

Logistic Regression shows higher false negatives, kNN has more scattered errors.
Random Forest achieves best balance overall.

Confusion Matrix - Random Forest Model



Confusion Matrix - kNN Model



Hyperparameter Search

01

RandomizedSearchCV

Efficient exploration of parameter space with 10 combinations tested

02

5-Fold Cross-Validation

Robust evaluation across different data splits

03

Parameter Space

Trees, depth, split criteria, leaf size explored systematically

04

Parallelized Execution

Completed in ~30 minutes with efficient computation

Optimization Results

Best Parameters Found

Optimal configuration improved baseline from 96.28% to 96.31% accuracy.

Small improvement validates good default parameters and confirms model robustness.

Insights from Tuning

- Default Random Forest parameters already strong
- Limited overfitting even with full depth
- Bootstrap sampling provides regularization
- Additional tuning could explore ensemble size

Computational Performance

180K	5min	15min	2sec
Dataset Size	Preprocessing Time	Training Time	Prediction Time
179,852 records × 32 features processed efficiently	Fast data cleaning and transformation	Random Forest model training duration	36K test samples—suitable for real-time deployment

Linear scaling with data size. Random Forest parallelizes across trees. Can handle monthly batches of millions of flights.

Deployment Architecture

Production Pipeline

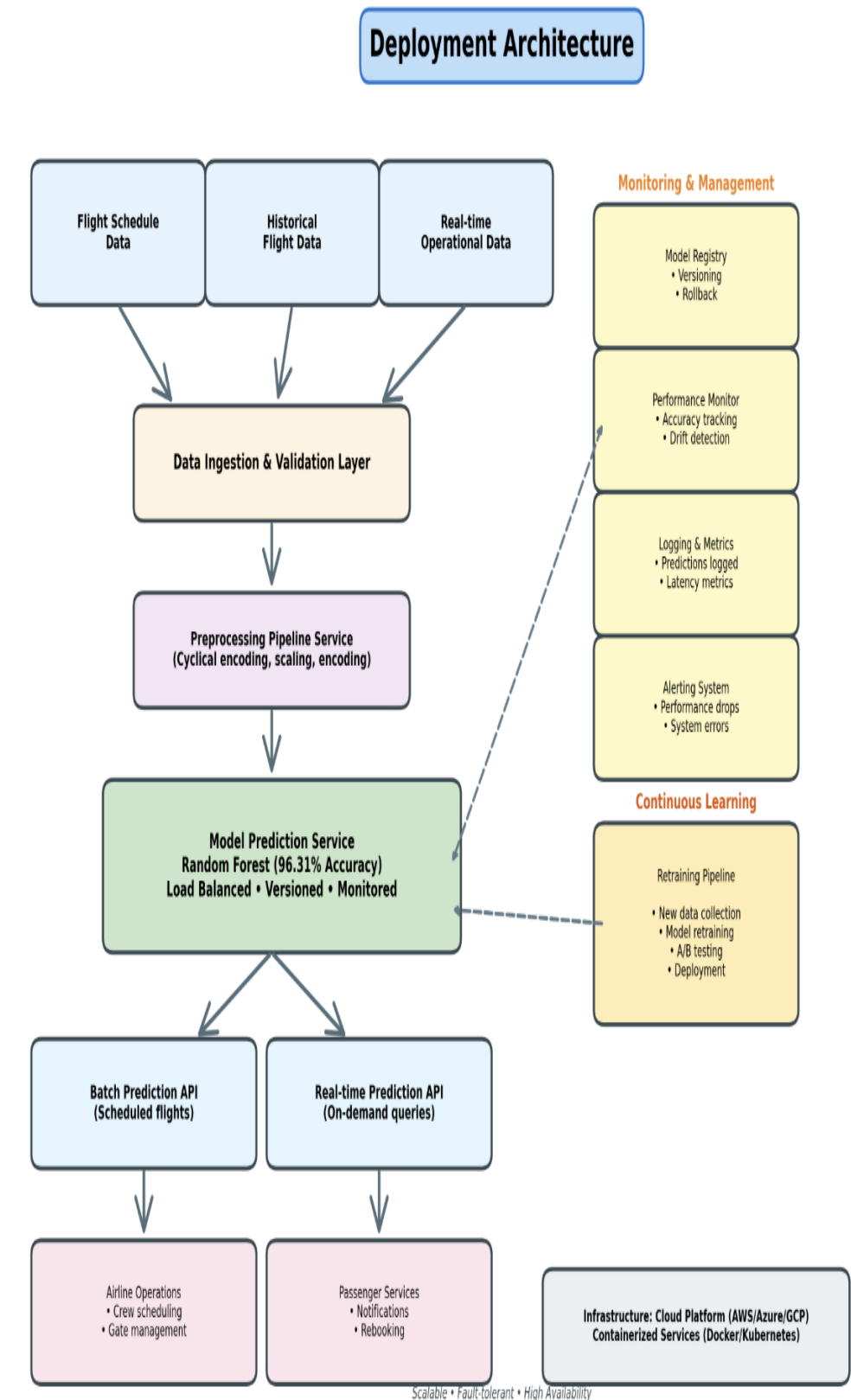
Batch prediction for scheduled flights with real-time API for on-demand queries

Model Management

Preprocessing pipeline serialization, model versioning, monitoring, and automated retraining

Integration Points

Airline operational systems, airport management platforms, passenger notifications, resource allocation



Current Limitations



Data Limitations

ATL-only training limits generalization. 2022 data only (single year). No external weather or air traffic control information. DepDel15 may not be available at prediction time.



Model Limitations

Binary classification loses magnitude information. 11% of delays still missed. Black-box nature limits interpretability. No explicit causality, only correlation.

Thank You

Yuntian Robin Wu Joshua Pina Dec. 4th, 2025